

60016 OPERATIONS RESEARCH

Finite Termination and Degeneracies

Lecture 05

Previously

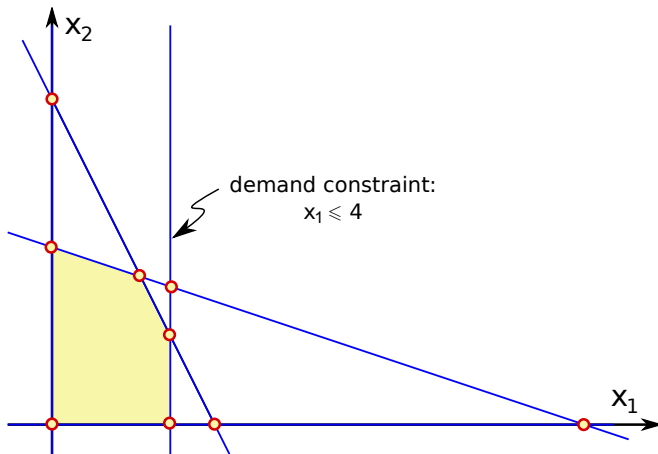
- ▶ Simplex tableau
- ▶ Pivoting
 - ▶ Pivoting equations
 - ▶ Pivot selection rules ensuring:
 - ▶ Non-inferiority
 - ▶ Feasibility

Now

- ▶ Finite termination in the Simplex method
- ▶ Degenerate BS's
- ▶ Finite termination theorem
- ▶ Cycling

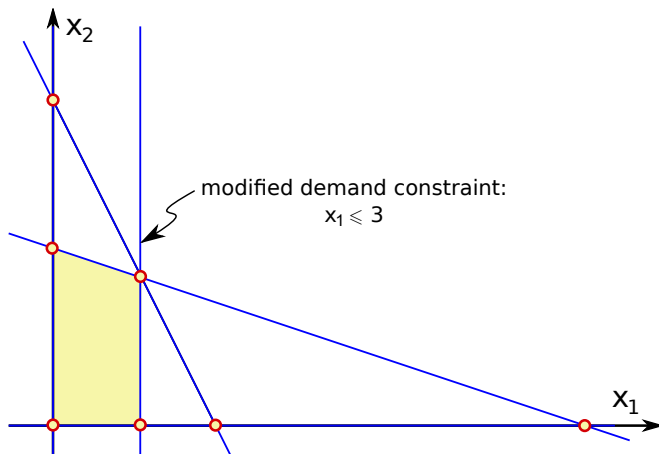
Degenerate BS's

Feasible set of Example 1:



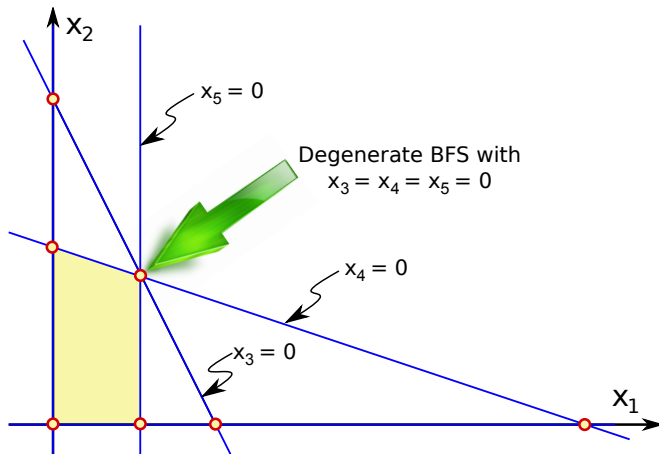
Degenerate BS's (cont)

Consider now a variant of Example 1:



Degenerate BS's (cont)

The highlighted BS corresponds to the index sets
 $I = \{1, 2, 3\}$, $I = \{1, 2, 4\}$ and $I = \{1, 2, 5\}$.



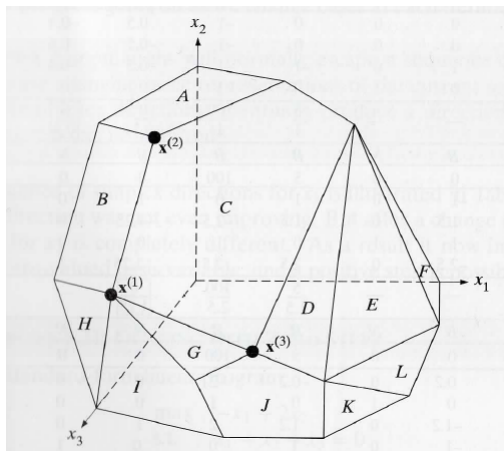
Degenerate BFS's: Definition

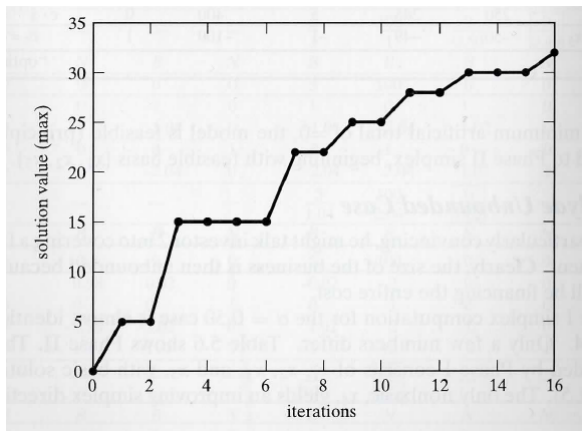
Definition: A BS is called **degenerate** if one or more basic variables (BVs) are zero.

$\Rightarrow$ A degenerate BS has more than $n - m$ zero-valued variables.

$\Rightarrow$ If we look at the tableau, there exists at least a BV such that $i \in I$ and $y_{i0} = 0$.

Definition: A BS is called **non-degenerate** if all of its basic variables are different from zero.





Finite Termination

Theorem: If all BFS's are non-degenerate, then the simplex algorithm must **terminate after a finite number of steps** with

- ▶ either an **optimal solution**
- ▶ or a proof that the problem is **unbounded**.

Finite Termination (Proof)

- ▶ At each step we have $y_{i0} > 0 \forall i \in I$ (non-degeneracy).
- ▶ Unless optimality or unboundedness is detected in STEP 1 or 2, we find $\beta'_0 = \beta_0 - \frac{\beta_q}{y_{pq}} y_{p0} < \beta_0$.
- ▶ Thus, the sequence of objective values obtained by the algorithm is strictly decreasing.

$$\beta_0 > \beta'_0 > \beta''_0 > \dots$$

No basic solution will ever be repeated!

- ▶ There are $\leq \binom{n}{m}$ basic solutions, since $\binom{n}{m}$ is the number of ways of picking m columns out of n to form an index set I .
- ▶ Thus, the process cannot continue indefinitely and must terminate at STEP 1 or 2 after a finite number of iterations (even though possibly a very large one!).



Degeneracy

Lemma: Assume that, $\forall i = 1, \dots, n, \exists \text{ BS } \hat{x} \text{ with } \hat{x}_i \neq 0$.

Then, a BS x is **degenerate** if and only if it is associated with **more than one index set**.

Proof: BS x degenerate $\Leftrightarrow$ BS x has more than one index set

- ▶ Suppose a BS x corresponds to index sets I_1 and I_2 , $I_1 \neq I_2$.
- ▶ Then $x_i = 0$ for all NBVs x_i with either $i \notin I_1$ or $i \notin I_2$ or both.
- ▶ In particular, since $I_1 \neq I_2$, there will be a NBV x_i in I_1 , that is a BV in I_2 . Since the two index sets describe the same BS x , x_i must be zero also in I_2 where it is basic.

$\Rightarrow$ x is a **degenerate BS**.

- ▶ (The same holds for any x_i that is NBV in I_2 and BV in I_1 .)

Degeneracy (cont)

Proof: BS x degenerate $\Rightarrow$ BS x has more than one index set

- ▶ Suppose x is a degenerate BS associated with some index set I ; consider the corresponding simplex tableau.
- ▶ Due to degeneracy, $\exists p \in I$ with $y_{p0} = 0$.
- ▶ $\exists q \notin I$ such that $y_{pq} \neq 0$. Otherwise, it would be always $x_p = 0$ everywhere which we assume impossible in the theorem statement.
- ▶ Pivoting on (p, q) gives a new basic solution which is identical to the current one since

$$y'_{q0} = \frac{y_{p0}}{y_{pq}} = 0 = y_{p0} \text{ and } y'_{i0} = y_{i0} - \frac{y_{iq}}{y_{pq}} y_{p0} = y_{i0} \quad \forall i \in I \setminus \{p\}.$$

$\Rightarrow x$ corresponds to the index sets I and $(I \setminus \{p\}) \cup \{q\}$.



Degeneracy and Simplex Algorithm

How does degeneracy affect the simplex algorithm?

- ▶ The index sets I and $(I \setminus \{p\}) \cup \{q\}$ produce the **same BFS** but **different basic representations**.
- ▶ If we pivot on (p, q) when $y_{p0} = 0$, then the **new BFS is identical to the old one**.
- ▶ In particular, we find

$$\beta'_0 = \beta_0 - \frac{\beta_q}{y_{pq}} y_{p0} = \beta_0,$$

and the **finite termination theorem breaks down** (no strict improvement of objective value).

- ▶ A pivot step (p, q) is called **degenerate** if $y_{p0} = 0$ and **non-degenerate** otherwise.

Degeneracy and Simplex Algorithm

The simplex algorithm can now be **decomposed** into:

$$\left[\begin{array}{c} \text{sequence of} \\ \text{degenerate} \\ \text{pivots} \end{array} \right] \quad \begin{array}{c} \text{non-} \\ \text{degenerate} \\ \text{pivot} \end{array} \quad \left[\begin{array}{c} \text{sequence of} \\ \text{degenerate} \\ \text{pivots} \end{array} \right] \dots$$

Note: Some or all of these sequences of degenerate pivots may be empty.

Geometrically, the current **BFS** remains **unchanged throughout a sequence of degenerate pivots**, and a non-degenerate pivot moves it to a different BFS.

Degeneracy and Simplex Algorithm

- ▶ We know that the number of index sets is $\leq \binom{n}{m}$.
- ⇒ Sequences of degenerate pivots are finite **if no index set is repeated**.
- ▶ However, on same rare instances pivoting can result in a cycling behaviour.
- ▶ In general, choosing degenerate pivots is a necessary condition, but not a sufficient one, for cycling.
- ▶ After a sequence of pivots, we return to the same index set and so the algorithm will **cycle and never terminate!**

Cycling Example

$$\min z = -\frac{3}{4}x_4 + 20x_5 - \frac{1}{2}x_6 + 6x_7$$

subject to:

$$\begin{array}{rccccrcrcl} x_1 & & +\frac{1}{4}x_4 & -8x_5 & -x_6 & +9x_7 & = & 0 \\ & x_2 & +\frac{1}{2}x_4 & -12x_5 & -\frac{1}{2}x_6 & +3x_7 & = & 0 \\ & & x_3 & & +x_6 & & = & 1 \end{array}$$

Use standard pivoting conventions!

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	0	$\frac{3}{4}$	-20	$\frac{1}{2}$	-6	0
x_1	1	0	0	$\frac{1}{4}$	-8	-1	9	0
x_2	0	1	0	$\frac{1}{2}$	-12	$-\frac{1}{2}$	3	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	0	0	$\frac{3}{4}$	-20	$\frac{1}{2}$	-6	0
x_1	1	0	0	$\frac{1}{4}$	-8	-1	9	0
x_2	0	1	0	$\frac{1}{2}$	-12	$-\frac{1}{2}$	3	0
x_3	0	0	1	0	0	1	0	1
z	-3	0	0	0	4	$\frac{7}{2}$	-33	0
x_4	4	0	0	1	-32	-4	36	0
x_2	-2	1	0	0	4	$\frac{3}{2}$	-15	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
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x_4	4	0	0	1	-32	-4	36	0
x_2	-2	1	0	0	4	$\frac{3}{2}$	-15	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

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x_4	4	0	0	1	-32	-4	36	0
x_2	-2	1	0	0	4	$\frac{3}{2}$	-15	0
x_3	0	0	1	0	0	1	0	1
z	-1	-1	0	0	0	2	-18	0
x_4	-12	8	0	1	0	8	-84	0
x_5	$-\frac{1}{2}$	$\frac{1}{4}$	0	0	1	$\frac{3}{8}$	$-\frac{15}{4}$	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	-1	-1	0	0	0	2	-18	0
x_4	-12	8	0	1	0	8	-84	0
x_5	$-\frac{1}{2}$	$\frac{1}{4}$	0	0	1	$\frac{3}{8}$	$-\frac{15}{4}$	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
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x_4	-12	8	0	1	0	8	-84	0
x_5	$-\frac{1}{2}$	$\frac{1}{4}$	0	0	1	$\frac{3}{8}$	$-\frac{15}{4}$	0
x_3	0	0	1	0	0	1	0	1
z	2	-3	0	$-\frac{1}{4}$	0	0	3	0
x_6	$-\frac{3}{2}$	1	0	$\frac{1}{8}$	0	1	$-\frac{21}{2}$	0
x_5	$\frac{1}{16}$	$-\frac{1}{8}$	0	$\frac{3}{64}$	1	0	$\frac{3}{16}$	0
x_3	$\frac{3}{2}$	-1	1	$-\frac{1}{8}$	0	0	$\frac{21}{2}$	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	2	-3	0	$-\frac{1}{4}$	0	0	3	0
x_6	$-\frac{3}{2}$	1	0	$\frac{1}{8}$	0	1	$-\frac{21}{2}$	0
x_5	$\frac{1}{16}$	$-\frac{1}{8}$	0	$\frac{3}{64}$	1	0	$\frac{3}{16}$	0
x_3	$\frac{3}{2}$	-1	1	$-\frac{1}{8}$	0	0	$\frac{21}{2}$	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	2	-3	0	$-\frac{1}{4}$	0	0	3	0
x_6	$-\frac{3}{2}$	1	0	$\frac{1}{8}$	0	1	$-\frac{21}{2}$	0
x_5	$\frac{1}{16}$	$-\frac{1}{8}$	0	$\frac{3}{64}$	1	0	$\frac{3}{16}$	0
x_3	$\frac{3}{2}$	-1	1	$-\frac{1}{8}$	0	0	$\frac{21}{2}$	1
z	1	-1	0	$\frac{1}{2}$	-16	0	0	0
x_6	2	-6	0	$-\frac{5}{2}$	56	1	0	0
x_7	$\frac{1}{3}$	$-\frac{2}{3}$	0	$-\frac{1}{4}$	$\frac{16}{3}$	0	1	0
x_3	-2	6	1	$\frac{5}{2}$	-56	0	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	-1	0	$\frac{1}{2}$	-16	0	0	0
x_6	2	-6	0	$-\frac{5}{2}$	56	1	0	0
x_7	$\frac{1}{3}$	$-\frac{2}{3}$	0	$-\frac{1}{4}$	$\frac{16}{3}$	0	1	0
x_3	-2	6	1	$\frac{5}{2}$	-56	0	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	-1	0	$\frac{1}{2}$	-16	0	0	0
x_6	2	-6	0	$-\frac{5}{2}$	56	1	0	0
x_7	$\frac{1}{3}$	$-\frac{2}{3}$	0	$-\frac{1}{4}$	$\frac{16}{3}$	0	1	0
x_3	-2	6	1	$\frac{5}{2}$	-56	0	0	1
z	0	2	0	$\frac{7}{4}$	-44	$-\frac{1}{2}$	0	0
x_1	1	-3	0	$-\frac{5}{4}$	28	$\frac{1}{2}$	0	0
x_7	0	$\frac{1}{3}$	0	$\frac{1}{6}$	-4	$-\frac{1}{6}$	1	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	2	0	$\frac{7}{4}$	-44	$-\frac{1}{2}$	0	0
x_1	1	-3	0	$-\frac{5}{4}$	28	$\frac{1}{2}$	0	0
x_7	0	$\frac{1}{3}$	0	$\frac{1}{6}$	-4	$-\frac{1}{6}$	1	0
x_3	0	0	1	0	0	1	0	1

Cycling Example (cont)

BV	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	0	2	0	$\frac{7}{4}$	-44	$-\frac{1}{2}$	0	0
x_1	1	-3	0	$-\frac{5}{4}$	28	$\frac{1}{2}$	0	0
x_7	0	$\frac{1}{3}$	0	$\frac{1}{6}$	-4	$-\frac{1}{6}$	1	0
x_3	0	0	1	0	0	1	0	1
z	0	0	0	$\frac{3}{4}$	-20	$\frac{1}{2}$	-6	0
x_1	1	0	0	$\frac{1}{4}$	-8	-1	9	0
x_2	0	1	0	$\frac{1}{2}$	-12	$-\frac{1}{2}$	3	0
x_3	0	0	1	0	0	1	0	1

This is the **initial basic representation** for $I = \{1, 2, 3\}$!

Bland's Rule

We can avoid cycling by amending the pivoting conventions.

Bland's Rule:

- (i) Choose the lowest-numbered (leftmost) nonbasic column q with a positive cost.

$$q = \min \{j \neq 0 \mid \beta_j > 0\}$$

- (ii) Denote as p the row with minimal $\bar{x}_{iq}$, in case of ties pick the row with the smallest index (same as standard conventions).

Theorem: With Bland's rule the simplex algorithm cannot cycle and hence is finite.

Degeneracy in Practice

- ▶ Cycling was thought to occur in **contrived examples**. For a long time it has therefore been **ignored in commercial solvers**.
- ▶ More recent experience with **larger and larger problems** indicates that cycling occurs, but it is still a **rare event**.
- ▶ Rigorous remedies such as **Bland's rule** are not satisfactory as they increase the **number of iterations** also in problems where cycling does not occur.
- ▶ In practice it is acceptable to **replace** a $y_{j0} = 0$ by $y_{j0} = \epsilon > 0$ (e.g., $\epsilon < 10^{-3}$) and then continue.